

## 2021년 1학기 C프로그래밍 3반 (40121-3)

### 실기시험2 풀이

5/27 (목요일) 13:00~14:50

아래 제시된 프로그램은 여러 가지 가능한 답 중 일부일 뿐이며, 요구되는 기능을 동일하게 수행하는 프로그램을 다른 방법으로 작성할 수 있다.

아래 프로그램 내용 중, 숙제10의 추가 문제 풀이에서 제시된 프로그램과 다른 부분(실질적으로 내용이 다른 부분을 의미함)은 그 라인 번호의 배경을 노란색으로 표시하였음

```
1  #include <stdio.h>
2  #include <string.h>
3  #include <stdlib.h>
4
5  #define MAX_WORD_COUNT 1000
6  #define MAX_WORD_LENGTH 101
7
8  #define MAX_LINE_LENGTH 2000
9
10 typedef struct wordEntry
11 {
12     char* word;
13     int count;
14 } Word;
15
16 int isSmallAlphabet(char c)
17 {
18     return ((c >= 'a' && c <= 'z') ? 1 : 0);
19 }
20
21 /*
22 단어 길이가 2이상이고
23 단어에 영어 알파벳이 하나라도 포함되어 있으면 유효한 단어임.
24 단, 영어 알파벳 소문자가 하나도 포함되어 있지 않은 것은 유효한 단어로 간주하지 않음
25 */
26 char* checkValidWord(char* p)
27 {
28     int i;
29
30     if (p == (char*)NULL || *p == '\0' || strlen(p) <= 1)
31         return NULL;
32
33     for (i = 0; p[i] != '\0'; i++)
34         if (isSmallAlphabet(p[i]))
35             return p;
36
37     return NULL;
38 }
39
40 int findWord(char* buf, Word* db, int wordCount)
41 {
42     int i;
43
44     if (wordCount == 0)
45         return -1;
46     for (i = 0; i < wordCount; i++)
47         if (strcmp(buf, db[i].word) == 0)
48             return i;
49     return -1;
50 }
51
52 int saveSingleWord(char* p, Word* db, int wordCount)
53 {
54     static char wordBuffer[MAX_WORD_COUNT][MAX_WORD_LENGTH];
55     static int wordBufferCount = 0;
```

```

56     int index;
57
58     if (p == (char*)NULL || *p == '\0')
59         return wordCount;
60
61     if ((index = findWord(p, db, wordCount)) < 0)
62         //the word not found in DB
63     {
64         db[wordCount].word = wordBuffer[wordBufferCount++];
65         strcpy(db[wordCount].word, p);
66         db[wordCount].count = 1;
67         wordCount++;
68     }
69     else // the word found in DB
70     {
71         db[index].count++;
72     }
73
74     return wordCount;
75 }
76
77 int getWords(Word* db, int maxNWords)
78 {
79     int wordCount = 0;
80     char buffer[MAX_LINE_LENGTH];
81     char* seps = " \t\n!\"#$%&'()*+,-./:;<=>?@[\\]^_`{|}~";
82     char* word;
83
84     while (fgets(buffer, MAX_LINE_LENGTH, stdin) != NULL)
85     {
86         if (strlen(buffer) == MAX_LINE_LENGTH - 1)
87             fprintf(stderr,
88                 "Warning: getWords: a long line with length %d detected!\n",
89                 strlen(buffer));
90
91         if ((word = strtok(buffer, seps)) != NULL)
92         {
93             if ((wordCount = saveSingleWord(
94                 checkValidWord(word), db, wordCount)) == maxNWords)
95             {
96                 fprintf(stderr,
97                     "Warning: getWords: max word count reached!\n");
98                 return(wordCount);
99             }
100             while ((word = strtok(NULL, seps)) != NULL)
101                 if ((wordCount = saveSingleWord(
102                     checkValidWord(word), db, wordCount)) == maxNWords)
103                 {
104                     fprintf(stderr,
105                         "Warning: getWords: max word count reached!\n");
106                     return(wordCount);
107                 }
108         }
109     }
110
111     return(wordCount);
112 }
113
114 void copyWordEntry(Word* dst, Word* src)
115 {
116     dst->word = src->word;
117     dst->count = src->count;
118 }
119
120 void swapWordEntry(Word* a, Word* b)
121 {
122     Word tmp;
123
124     copyWordEntry(&tmp, a);
125     copyWordEntry(a, b);
126     copyWordEntry(b, &tmp);
127 }

```

```

128
129 /* perform bubble sort by words in dictionary order */
130 void sortWords(Word* db, int n)
131 {
132     int i, j;
133
134     for (i = 0; i < n - 1; i++)
135         for (j = 0; j < (n - i) - 1; j++)
136             if (strcmp(db[j].word, db[j + 1].word) > 0)
137                 swapWordEntry(&db[j], &db[j + 1]);
138 }
139
140 /* perform bubble sort by word length in decreasing order */
141 void sortWordsByLength(Word* db, int n)
142 {
143     int i, j;
144
145     for (i = 0; i < n - 1; i++)
146         for (j = 0; j < (n - i) - 1; j++)
147             if (strlen(db[j].word) < strlen(db[j + 1].word))
148                 swapWordEntry(&db[j], &db[j + 1]);
149 }
150
151 void putWords(Word* db, int wordCount)
152 {
153     int i;
154
155     printf("Number of different words = %d\n", wordCount);
156     for (i = 0; i < wordCount; i++)
157         printf("%04d: %4d %3d %s\n",
158             i + 1, db[i].count, strlen(db[i].word), db[i].word);
159 }
160
161 int main(void)
162 {
163     Word allWords[MAX_WORD_COUNT] = { 0 };
164     int wordCount, idx, cnt;
165
166     wordCount = getWords(allWords, MAX_WORD_COUNT);
167
168     // Sort the words by their lengths in decreasing order
169     sortWordsByLength(allWords, wordCount);
170
171     // For each group of words with equal lengths
172     for (idx = 0; idx < wordCount; idx += cnt)
173     {
174         // Find the boundary of the next group of words
175         // with equal length
176         for (cnt = 1; idx + cnt < wordCount; cnt++)
177             if (strlen(allWords[idx + cnt].word) !=
178                 strlen(allWords[idx].word))
179                 break;
180
181         // Sort the words in each subgroup of words
182         // with equal lengths in dictionary order
183         if (cnt > 1)
184             sortWords(&allWords[idx], cnt);
185     }
186
187     putWords(allWords, wordCount);
188
189     fprintf(stderr, "Processing completed!!\n"); return 0;
190 }

```

채점용 입력 파일: biden.txt (숙제9 및 숙제10의 입력 파일과 동일. 편하게 참조할 수 있게 이 폴이에도 포함)

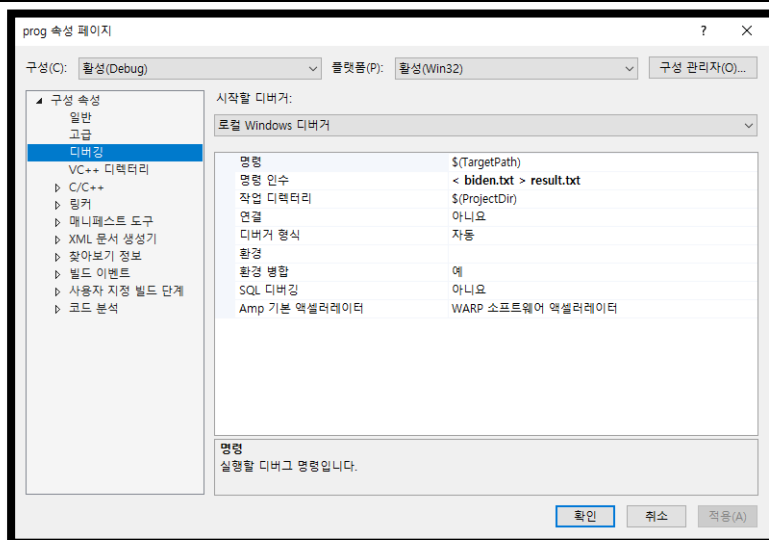
Joseph Robinette Biden Jr. (born November 20, 1942) is an American politician who is the 46th president of the United States. A member of the Democratic Party, he served as the 47th vice president from 2009 to 2017 under Barack Obama and represented Delaware in the United States Senate from 1973 to 2009.

Born and raised in Scranton, Pennsylvania, and later in New Castle County, Delaware, Biden studied at the University of Delaware before earning his law degree from Syracuse University in 1968. He was elected to the New Castle County Council in 1970 and became the sixth-youngest senator ever when he was elected to the U.S. Senate from Delaware in 1972, at age 29. Biden was a longtime member of the Senate Foreign Relations Committee and eventually became its chairman. He also chaired the Senate Judiciary Committee from 1987 to 1995, dealing with drug policy, crime prevention, and civil liberties issues; led the effort to pass the Violent Crime Control and Law Enforcement Act and the Violence Against Women Act; and oversaw six U.S. Supreme Court confirmation hearings, including the contentious hearings for Robert Bork and Clarence Thomas. He ran unsuccessfully for the Democratic presidential nomination in 1988 and 2008. Biden was reelected to the Senate six times, and was the fourth-most senior senator when he became Obama's vice president after they won the 2008 presidential election.

During eight years as vice president, Biden leaned on his Senate experience and frequently represented the administration in negotiations with congressional Republicans, including on the Budget Control Act of 2011, which resolved a debt ceiling crisis, and the American Taxpayer Relief Act of 2012, which addressed the impending "fiscal cliff". He also oversaw infrastructure spending in 2009 to counteract the Great Recession. On foreign policy, Biden was a close counselor to the president and took a leading role in designing the withdrawal of U.S. troops from Iraq in 2011. In 2017, Obama awarded Biden the Presidential Medal of Freedom with distinction, making him the first president to receive it before taking office.

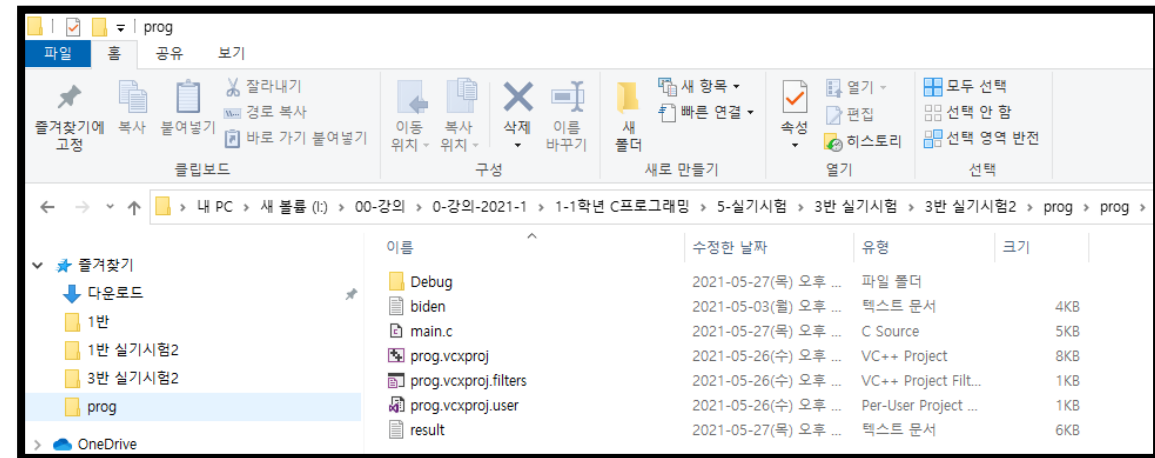
On April 25, 2019, Biden announced his candidacy in the 2020 presidential election. He became the presumptive Democratic nominee in April 2020 and reached the delegate threshold needed to secure the nomination in June 2020. Biden and his running mate Kamala Harris defeated incumbent president Donald Trump and vice president Mike Pence in the general election. Biden is the oldest elected president, the first from Delaware, and the second Catholic. His early presidential activity centered around proposing, lobbying for, and signing into law the American Rescue Plan Act of 2021 to speed up the United States' recovery from the COVID-19 pandemic and the ongoing recession, as well as a series of executive orders. Biden's orders addressed the pandemic and reversed several Trump administration policies, including by rejoining the Paris Agreement on climate change and reaffirming protections for Deferred Action for Childhood Arrivals recipients. In April 2021, Biden announced the withdrawal of all U.S. troops from Afghanistan by September 2021.

아래는 VS2019 내에서 실행할 때, 명령 인수 속성 설정 내용임

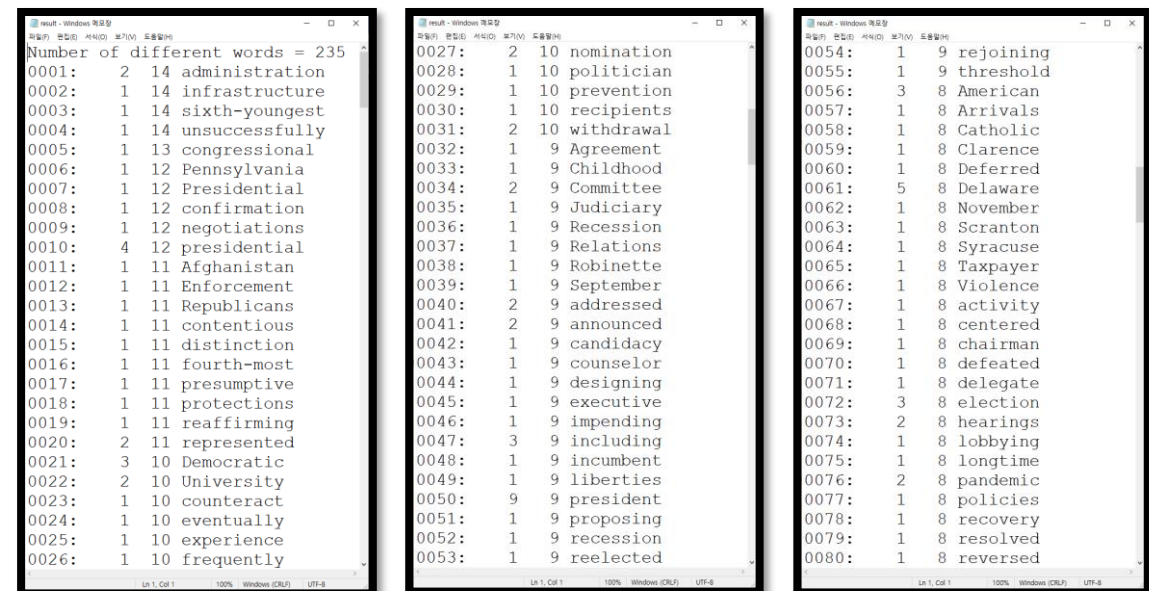


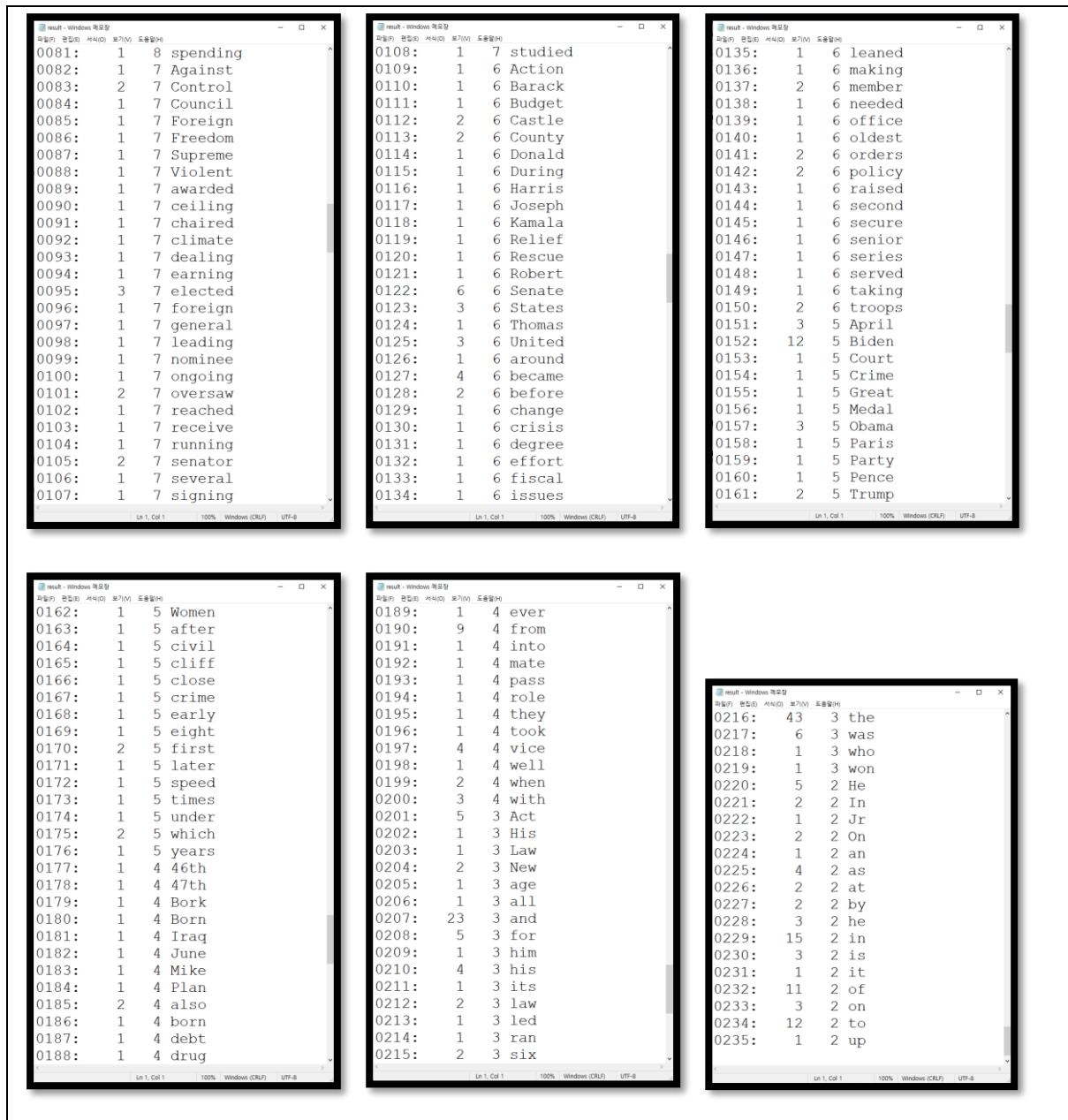
VS2019 내에서 실행할 때, 입력 리다이렉션에 의한 입력 파일은 작업 디렉토리 즉 해당 프로젝트의 프로젝트 폴더 내에

두어야 하고, 출력 리다이렉션에 의한 출력 파일도 프로젝트 폴더 내에 생성됨. 아래는 프로그램을 한 차례 실행한 후, 파일 탐색기를 사용하여 프로젝트 폴더의 내용을 보여준 것임



**출력 파일:** 특히 숙제9 또는 숙제10의 결과와 달리 “COVID-19”가 유효한 단어로 간주되지 않았음에 주목.





끝.